Md Ahsanul Haque, Ph.D.
MAT-120

Quiz-3

Date: April 19, 2026

Name

Section

Instructions:

- Write your full name and section number.
- This quiz contains 2 problems. It is **YOUR responsibility** to check that your copy has all 2 problems.
- Make sure to read the instructions for each problem carefully.
- Use correct mathematical notation.
- Write cohesively and follow a logical sequence of steps.
- Check that your answer can be read and understood by the grader.

1. (10 points) Consider the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$.
   (a) Determine the intervals where $f(x)$ is increasing and decreasing.
   (b) Determine the intervals where $f(x)$ is concave up and concave down.
   (c) Find all relative extrema.
   (d) Find all points of inflection.

2. (10 points) Find the absolute maximum and absolute minimum values of the function:

$$g(x) = \frac{2x^2 - 8}{x^2 + 3}$$

on the closed interval $[-2, 4]$.